

```
for object to mirror...
mirror_mod.mirror_object

operation == "MIRROR_X":
    mirror_mod.use_x = True
    mirror_mod.use_y = False
    mirror_mod.use_z = False
operation == "MIRROR_Y":
    mirror_mod.use_x = False
    mirror_mod.use_y = True
    mirror_mod.use_z = False
operation == "MIRROR_Z":
    mirror_mod.use_x = False
    mirror_mod.use_y = False
    mirror_mod.use_z = True

#selection at the end -add
mirror_ob.select= 1
mirror_ob.select=1
context.scene.objects.active
("Selected" + str(modifier_...
mirror_ob.select = 0
= bpy.context.selected_object
data.objects[one.name].select

print("please select exactly

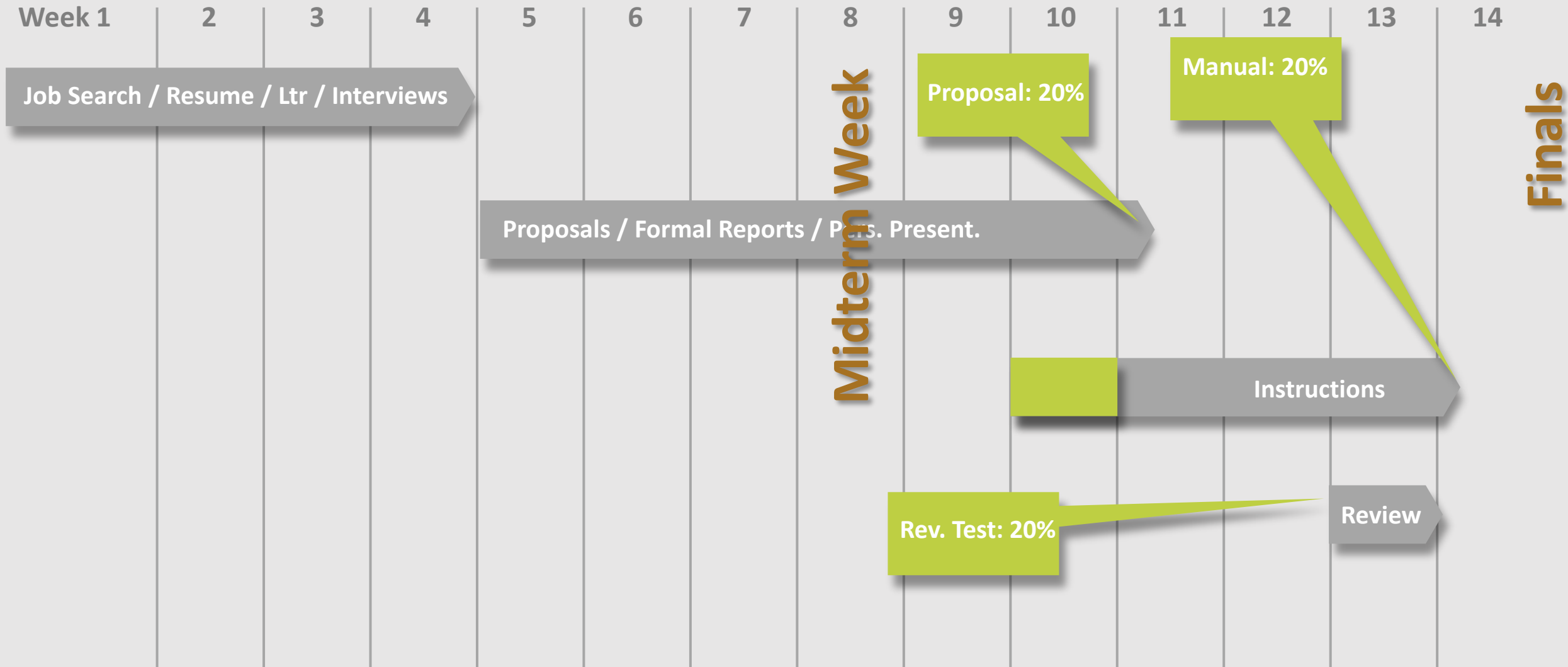
-- OPERATOR CLASSES -----

types.Operator):
    X mirror to the selected
    object.mirror_mirror_x"
    mirror X"
```

Comm 2216

Week 10:
Instructions

Course Schedule



Instructions

Lecture 1: (today) describes the overall purpose and organization of instructions to get you started with the next assignment.

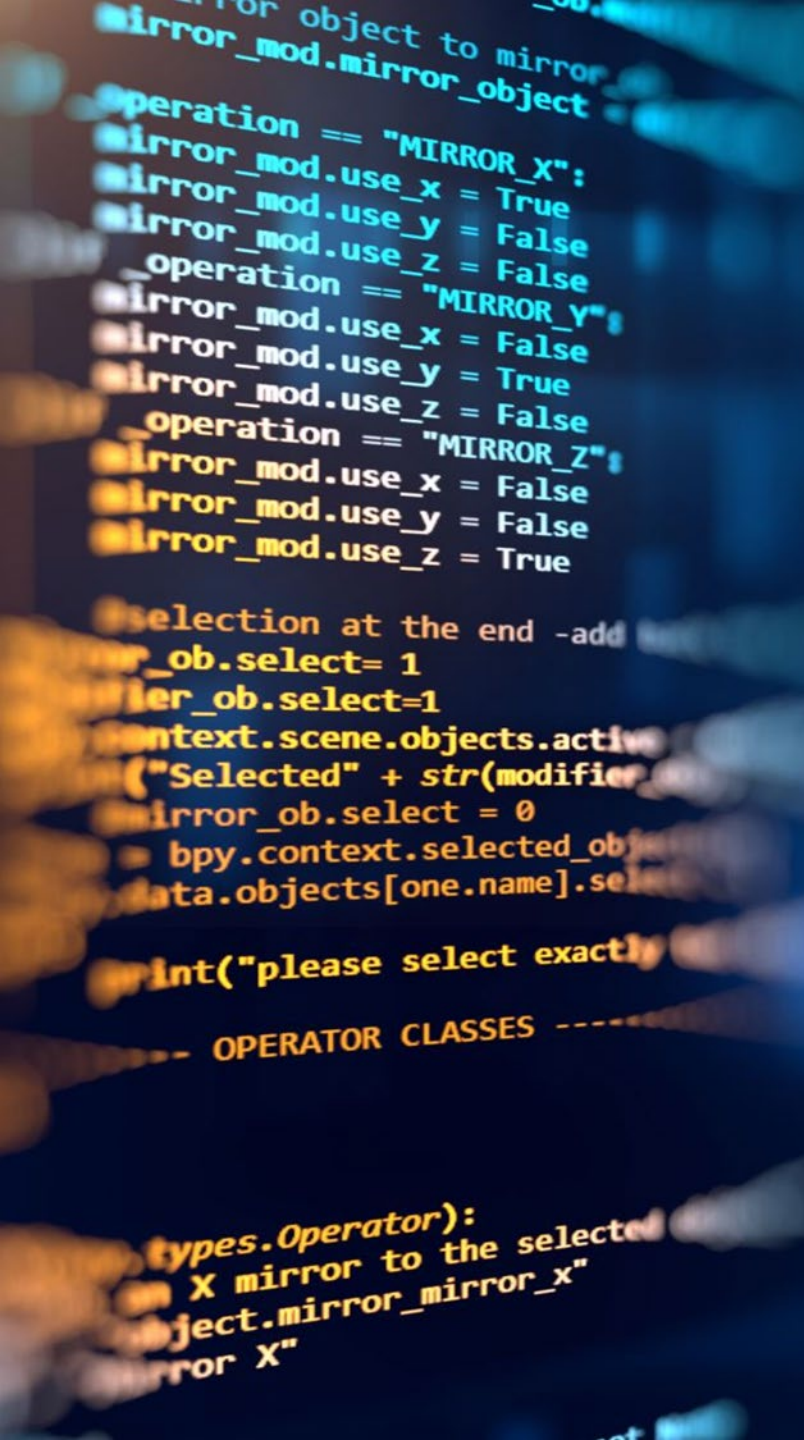
Lecture 2: will detail the structure of manuals and how to format them and to annotate the screen captures.

Instructions, parte uno

What is the assignment?

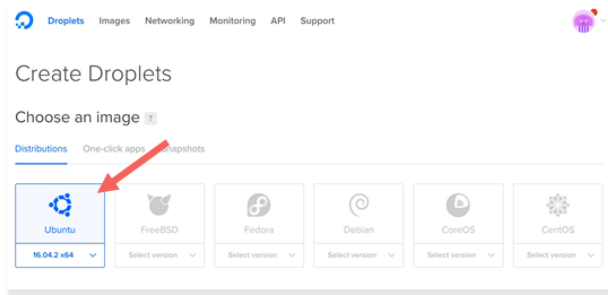
Why are you doing this to us?

How do we write instructions?



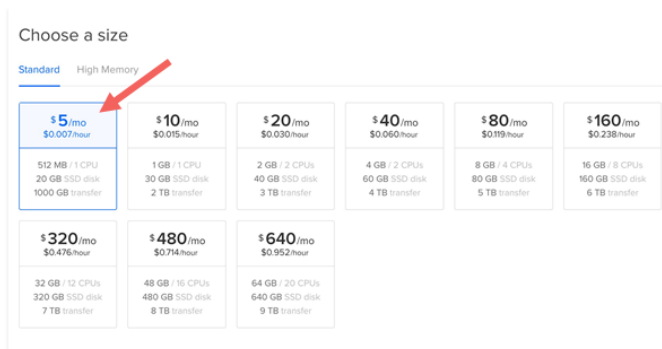
Note: Ubuntu is the operating system that the server will run on. It is free to use, and one of the most commonly used operating systems to run servers.

3.) Select *Ubuntu* under the distributions tab.



Warning: If you do not choose the \$5/month option, you could be billed a larger amount for the server.

4.) Scroll down to the *Choose a size* option and select the **\$5_{mo}** size.



1. The Assignment

- You will get together in self-selected pairs or groups of three to document one of your own apps
- You will write a 12-page (min.) instruction manual with an intro, the instruction sets, a glossary, and a conclusion/troubleshooting section.
- A considerable part of the real estate on the pages will be taken up by screen captures.
- The assignment is worth 20% of the course grade.
- The assignment handout, with checklist, is on D2L, Week 10.
- The sign-up sheet is linked on D2L, Week 10 as well.
- Sample manuals are in Week 11.

1. The Assignment, cont'd

The following is the timeframe we're looking at:

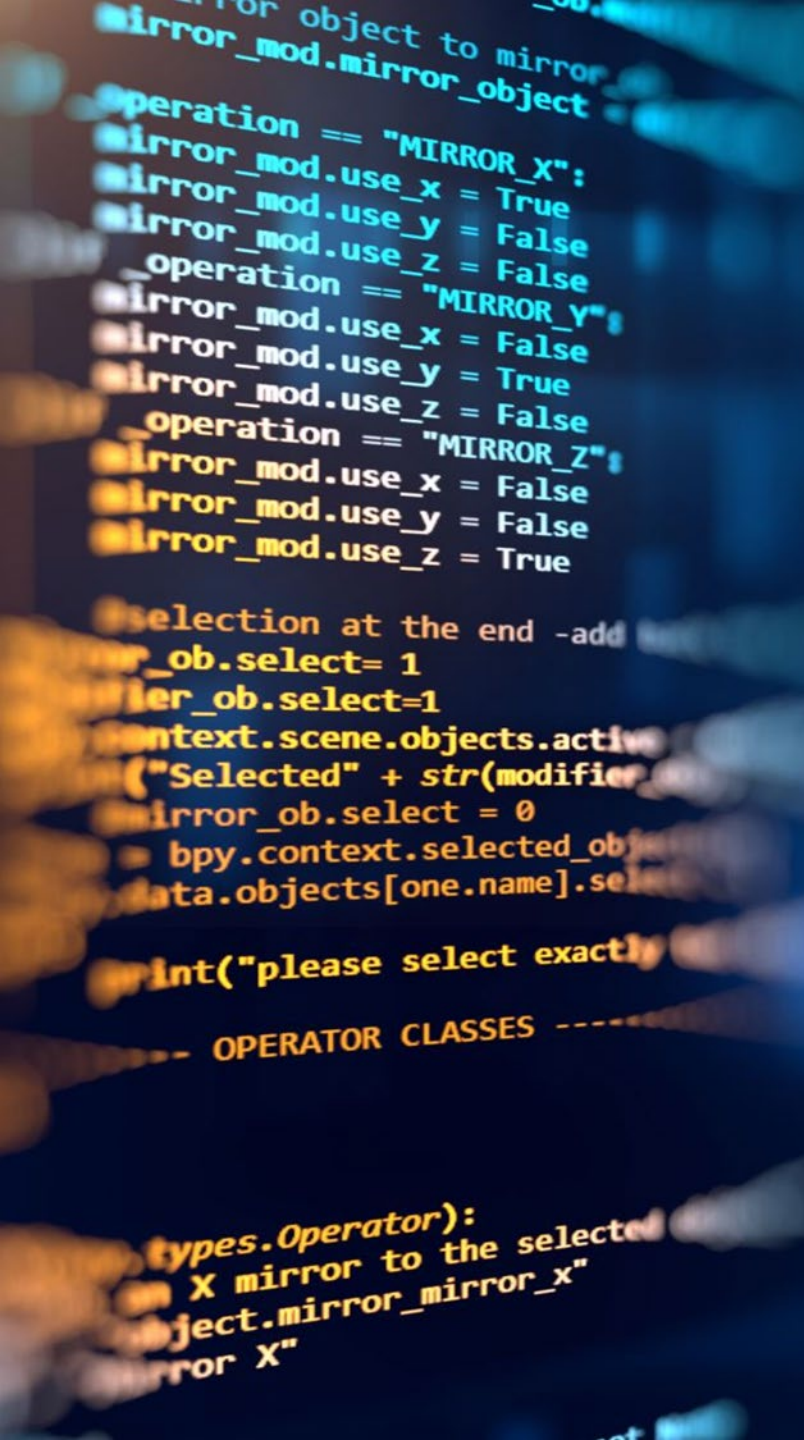
- Lab of Week 11 (Nov. 10): teams picked, topic chosen, link submitted, outline started
- Lab of Week 12 (Nov. 17): outline completed, work started
- Week 13 (Nov. 24): easy, breezy Term-end Review Test (20 gentle marks). Completed in the labs. Details to follow.
- Tuesday of Week 14 (Dec. 03, midnight) the manual is due.

Instructions, parte uno

What is the assignment?

Why are you doing this to us?

How do we write instructions?





2. Why?!

Because coders and programmers are frequently asked to document their products. Not all companies have or engage professional technical writers out of considerations for cost or timelines.

Frequently documentation gets produced in parallel with the product and is subject to last minute changes before release (or needs quick fixes after release).

CST_{ears}



Who better to do this than you, especially if you are the low person on the totem pole?

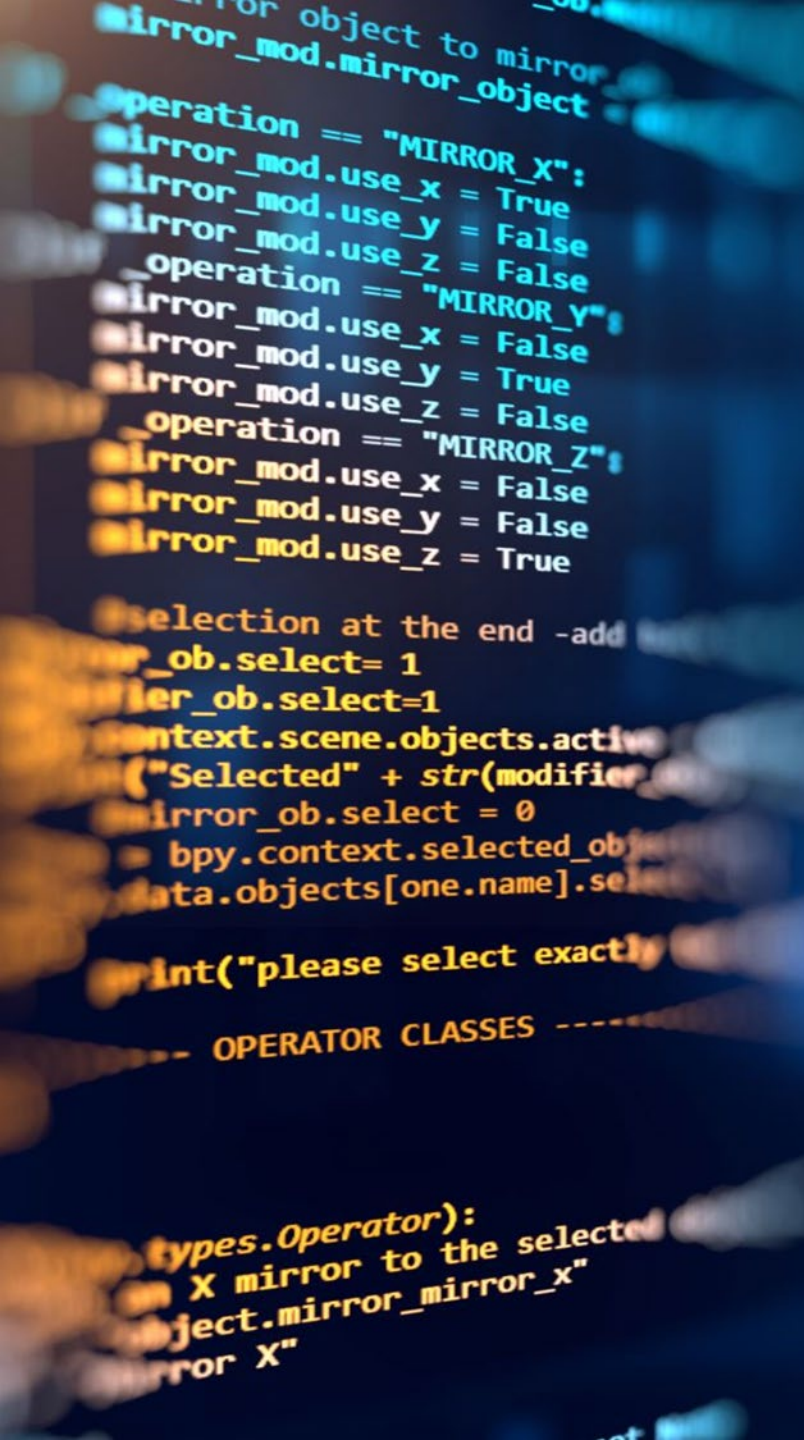
You are going to have to get used to documenting your work.

Instructions, parte uno

What is the assignment?

Why are you doing this to us?

How do we write instructions?



Purpose of instructions

Instructions do not describe processes or explain operational principles.

Surprisingly, they are not meant to make people understand the software or process.

The only purpose of instructions is to tell the reader's hands and eyes what to do in order to complete a task.

It's useful to think of instructions as a sort of programming language for users.

Alternatively, it's useful to think of the user as a tool or robot to get the job done. You need to program that robot.



What do we know about the user?

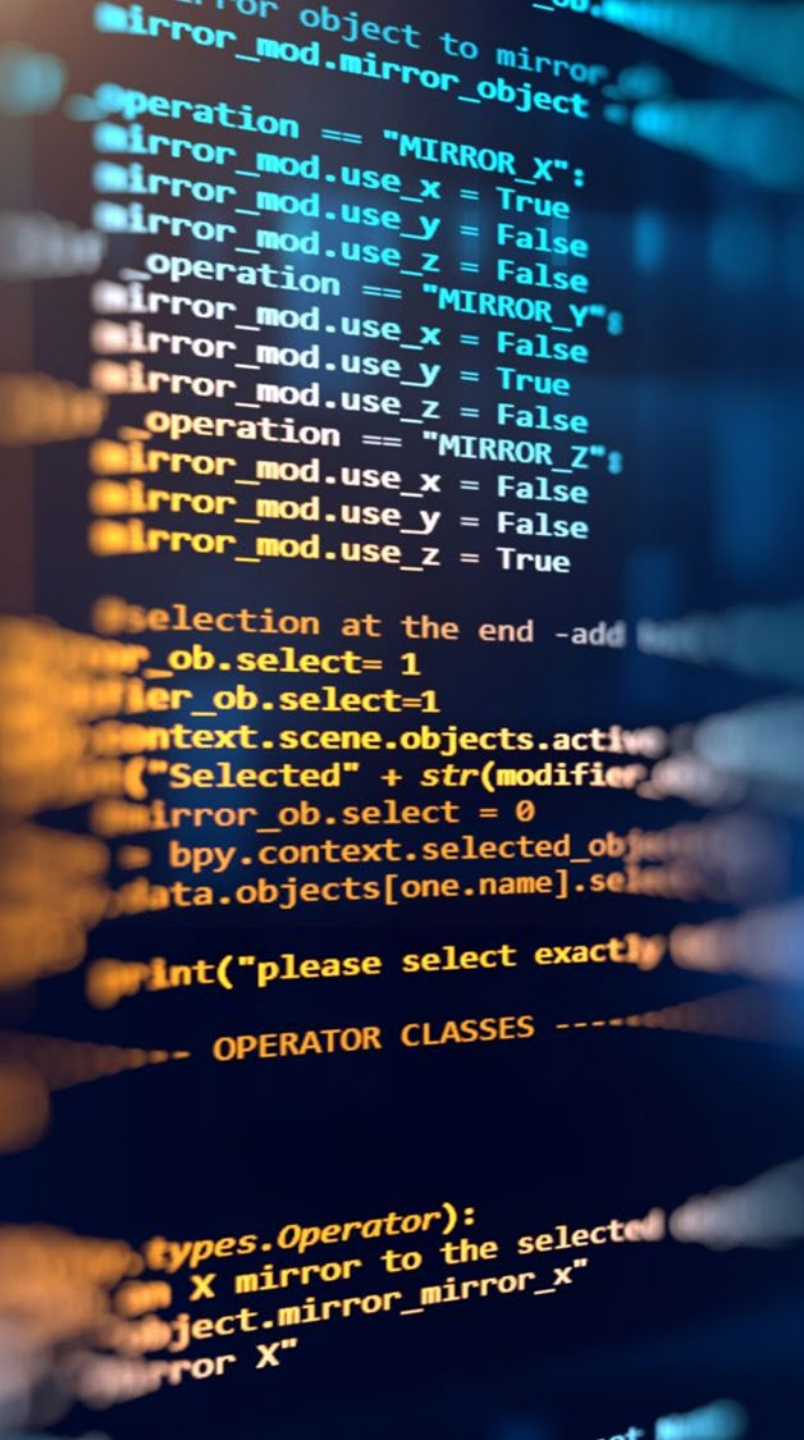
Users will not read the full instructions before starting the procedure.

They will at best scan for whatever looks cool or jumps out at them.

They will not remember information from step to step or understand underlying principles.

From this follows the cardinal rule for instructions:

Use a strictly chronological order and spoon-feed (or trickle-feed) information to the user as it becomes relevant.

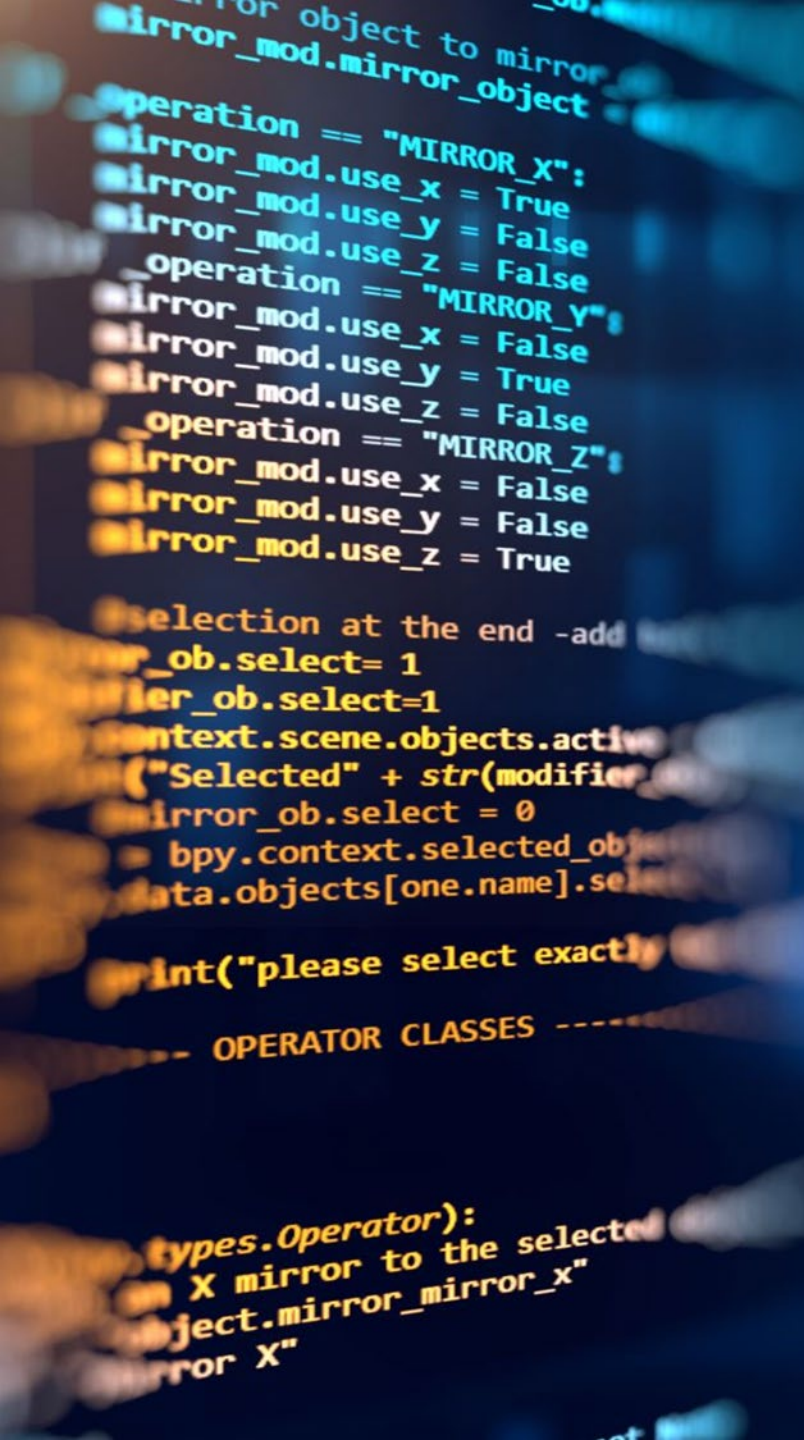


Some terminology

Instructions are the sequence of steps to perform a task; a **procedure** is the whole written task or the series of separate instruction sets that make up a complex task.

The three parts of a **procedure** are the introduction, the step-by-step instructions, and the conclusion (generally with a trouble-shooting guide, occasionally with a glossary).

A **manual** brings together all the tasks (procedures) associated with an app or a mechanism (such as a bicycle, car, plane, software, or app)



Structure of Procedures

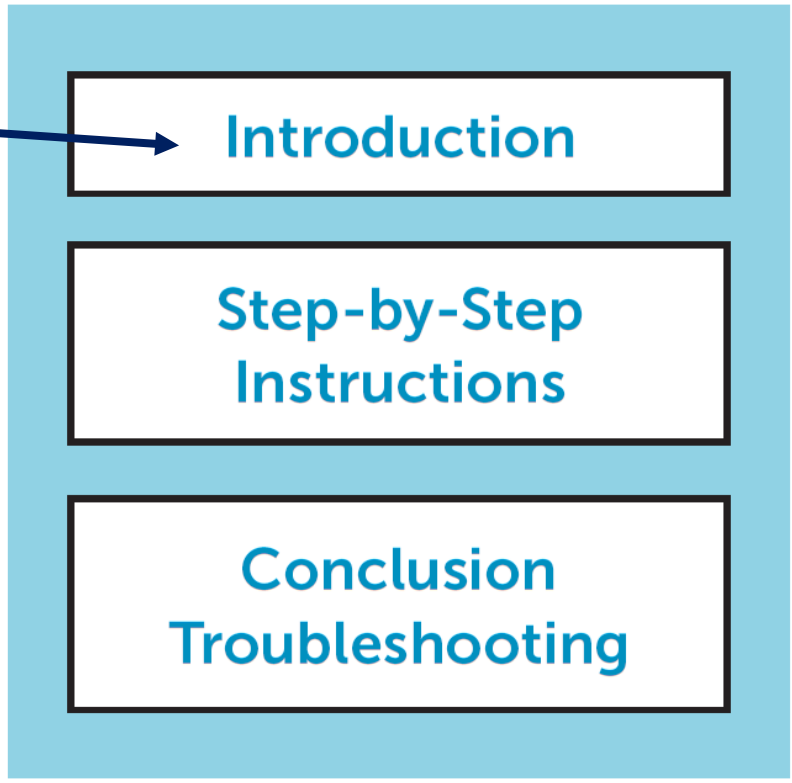
How does the procedure fit into the larger context and what is the benefit of performing the procedure?

When or under what circumstances should the procedure be followed and by whom?

What tools and materials will be required and how much time should it take?

What notes, cautions, warnings, and danger alerts apply to the procedure as a whole?

What naming and typographical conventions will be used in the procedure?

A diagram showing the structure of a procedure. It consists of a light blue rectangular container with a thin white border. Inside this container are three white rectangular boxes, each with a thin black border, stacked vertically. The top box contains the word 'Introduction' in blue text. The middle box contains the words 'Step-by-Step' and 'Instructions' in blue text, with 'Instructions' on a new line. The bottom box contains the words 'Conclusion' and 'Troubleshooting' in blue text, with 'Troubleshooting' on a new line. A dark blue arrow points from the text 'How does the procedure fit into the larger context and what is the benefit of performing the procedure?' to the 'Introduction' box.

Introduction

Step-by-Step
Instructions

Conclusion
Troubleshooting

Introduction

These instructions are on how to create, connect, and upload your first website using the online application Digital Ocean. The application makes hosting a web server super easy. It's used by students and industry professionals every day because of its cheap prices, and basic user interface.

This manual is for users with some basic server administration skills, so they can learn how to navigate and use the Digital Ocean platform.

For this tutorial, we will be using the most recent version of the Digital Ocean application. The website requires you to use the most recent version, so no need to worry about which version you use.

Prerequisites

To follow these instructions, you will need:

- Mac Operating System.
- DigitalOcean.com Account.
- FileZilla File Upload Application.
- Basic server administration skills.
- Website Code Files.

CONVENTIONS

These instructions use the following typographic instructions:

[transit] > [options]
directions*

Start

Menu Selection Sequences

Glossary word

Clickable buttons

Task One – Creating Your Server

The first step to hosting a website is creating the server that will store the sites files. Currently, the best server for small websites is the \$5/month plan on Digital Ocean. We will show you how to setup this server in a few simple steps.

Structure of Procedures, cont'd

What specific actions should the reader's hands perform and in what specific sequence?

What notes, cautions, warnings, and danger alerts apply to these individual steps?

Warning: If you do not choose the \$5/month option, you could be billed a larger amount for the server.

4.) Scroll down to the *Choose a size* option and select the ~~\$5~~_{5/mo} size.

Introduction

Step-by-Step
Instructions

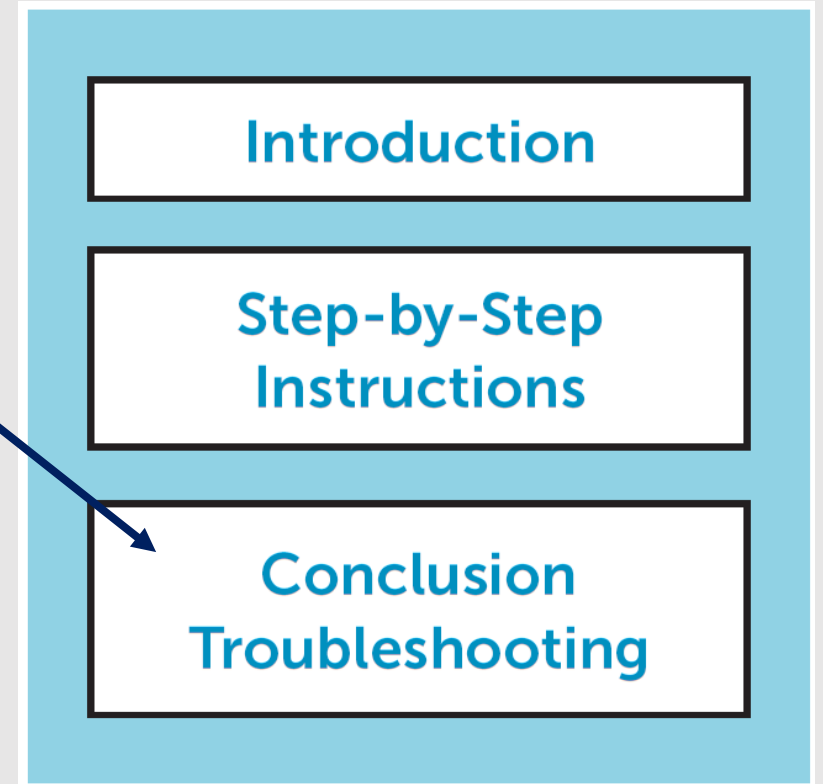
Conclusion
Troubleshooting

Structure of Procedures, cont'd

Conclusion: what quick test will let the user know that the procedure has been followed correctly?

Troubleshooting Guide: when users make predictable (or frequent) errors in a procedure, how can those common problems be fixed?

Glossary: what terms that may be unfamiliar to the reader, are being used in the procedures and what do they mean?
Place at the end of the manual.



Conclusion

Once you have uploaded your website files onto the server, you have completed the tutorial. If you visit your server IP in a browser, you can see your website live on the web. Visit `http://<your_server_ip>` and you can see and share your website.

With the skills learnt from this tutorial, you can now quickly start and host websites using the Digital Ocean platform.

Troubleshooting Guide

Can't Connect To Server.

Check Digital Oceans twitter account or your email address for any downtime at the service. This is a typical occurrence at some of their datacenters around the country. Link to Digital Ocean uptime status: twitter.com/dostatus

Forgot Password?

If you forget your password, visit the [Access tab](#) and press the *Reset Root Password* button for a new server password.

Glossary

Dispute: A formal claim filed when players disagree about a match outcome. These can be adjudicated by . . .

Entry Fee: The amount paid by players to join a match

Wager Match: A competitive match where players bet money on their performance.

Glossary

Droplet

Droplets are Digital Ocean's definition of a server. A virtual server is a server that shares hardware and software resources with other operating systems.

IP Address

Internet Protocol Address is a unique identifier assigned to each computer and other device connected to a network.

Step-by-step Instructions

Use the following rules to make the step-by-step instructions as clear as possible:

- Number all the steps
- Begin every step with a verb in the imperative mood
- Make sure the verb describes the action that drives the process.
- Begin with an adverb if it is important how the action is performed

~~Step 17.~~ Attach the hose to the drain valve.

~~Step 23)~~ Carefully remove the detonator

~~Step 2:~~ Calmly and slowly back away from the bear sow and her cub.

Not: Go to the top right corner and click **START**.

But: Click **START** in the top right corner.

Step-by-step Instructions

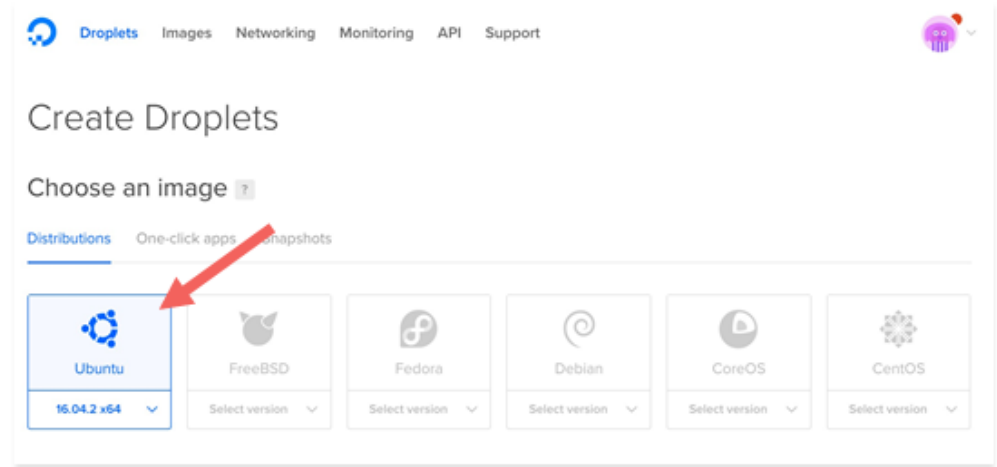
Include graphics wherever you can.

Otherwise, use “locate” steps for components that are difficult to find (generally doesn’t apply to software instructions).

Subdivide long instruction sets into several sections: provide clear descriptive headings.

Note: Ubuntu is the operating system that the server will run on. It is free to use, and one of the most commonly used operating systems to run servers.

3.) Select *Ubuntu* under the distributions tab.



Conclusions to procedures

First-time readers might wonder whether they've performed the procedure correctly.

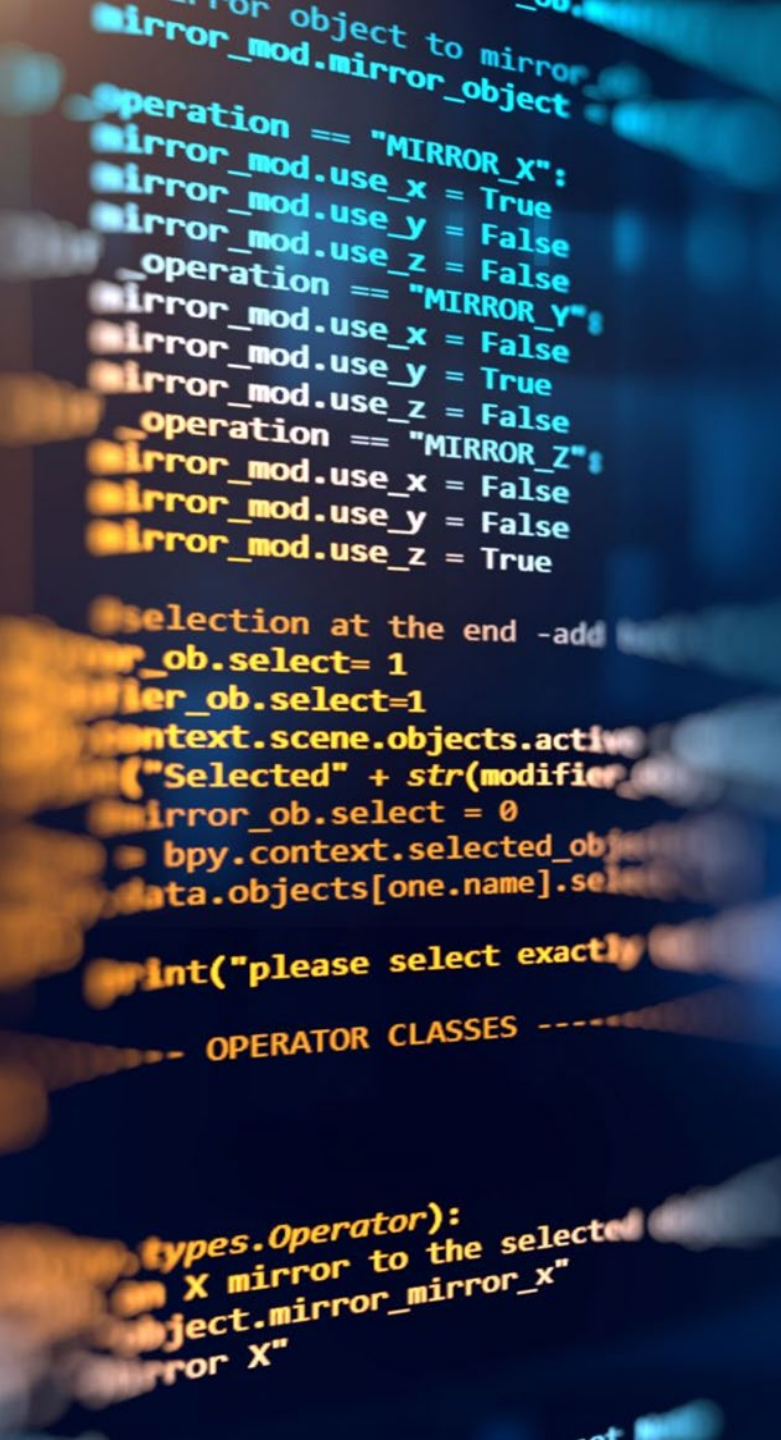
Provide a quick check to reassure them.

All three status lights on the modem should be steady green

If readers (can) make common mistakes, add a troubleshooting guide.

If one of the status lights is flashing amber, check that the ...

If you are using terminology with which the reader may not be familiar, provide a glossary.



Notes, Cautions, Warnings

Place notes, cautions, warnings, and danger alerts immediately before the step to which they apply.

This is part of the strictly chronological approach to delivering information.

Failure to provide adequate and correctly placed cautions, notes, and warnings can create liability risks.

If the notes apply to the entire procedure, place them in the introduction.

Note: the notes, cautions, and warnings do not take the place of the instruction step.

Warning: If you do not choose the \$5/month option, you could be billed a larger amount for the server.

4.) Scroll down to the *Choose a size* option and select the \$5_{max} size.

Notes, Cautions, Warnings

Example:

Note: If the screws are tightened fully at this point the frame will become inflexible. This will make it difficult to fit in the cross brace in Step 19.

14. Screw the support bracket to the back brace.

Notes provide merely additional, helpful, not critical information

They may be included in the instruction step.

14. Screw the support bracket to the back brace. Leave the screw loose at this point to make it easier to fit in the cross brace in Step 19.

Note: the notes, cautions, and warnings do not take the place of the instruction step.

Notes, Cautions, Warnings

Notes, cautions, and warnings should be formatted with increasing emphasis.

Readers should be in no doubt as to the threat level.

Depending on the hazards the reader is likely to encounter, two or three threat levels may be necessary.

```
mirror object to mirror  
mirror_mod.mirror_object
```

```
operation == "MIRROR_X":  
    mirror_mod.use_x = True  
    mirror_mod.use_y = False  
    mirror_mod.use_z = False  
operation == "MIRROR_Y":  
    mirror_mod.use_x = False  
    mirror_mod.use_y = True  
    mirror_mod.use_z = False  
operation == "MIRROR_Z":  
    mirror_mod.use_x = False  
    mirror_mod.use_y = False  
    mirror_mod.use_z = True
```

```
#selection at the end -add  
mirror_ob.select= 1  
mirror_ob.select=1  
context.scene.objects.active  
("Selected" + str(modifier))  
mirror_ob.select = 0  
= bpy.context.selected_object  
data.objects[one.name].select
```

```
print("please select exactly
```

```
-- OPERATOR CLASSES -----
```

```
types.Operator):  
    X mirror to the selected  
    object.mirror_mirror_x"  
    mirror X"
```



```
...mirror object to mirror_...  
mirror_mod.mirror_object = ...  
...operation == "MIRROR X":  
mirror_mod.use_x = False  
mirror_mod.use_y = True  
mirror_mod.use_z = False  
...operation == "MIRROR_Z":  
mirror_mod.use_x = True  
mirror_mod.use_y = False  
mirror_mod.use_z = True  
...  
...select ...  
...mirror_ob.select=1  
...context.scene.objects.active  
...("Selected" + str(modifier...  
...mirror_ob.select = 0  
...= bpy.context.selected_object  
...data.objects[one.name].select  
...  
...print("...test exactly...  
...  
...-- OPERATOR CLASSES
```

Notes make steps easier to follow

Cautions prevent failure of the procedure or damage to equipment

Warnings prevent injury

```
...types  
...X m  
...object  
...mirror X
```

Danger alerts prevent potential death



Note: Two different-sized screws are provided. Be sure to use the 3/8 in. screw here.



Caution: Do not use non-rechargeable batteries in this charging unit or the charging unit could be damaged.



Warning: Put on safety goggles and a breathing mask before removing the battery caps. Make sure the workspace is well ventilated.



DANGER: EXTREME ELECTROCUTION HAZARD. DO NOT ATTEMPT THIS STEP WITHOUT VERIFYING THAT PROPER LOCKOUT PROCEDURES HAVE BEEN FOLLOWED.


```
object to mirror  
_mod.mirror_object  
ion == "MIRROR_X":  
_mod.use_x = True  
_mod.use_y = False  
_mod.use_z = False  
ation == "MIRROR_Y":  
_mod.use_x = False  
_mod.use_y = True  
_mod.use_z = False  
ation == "MIRROR_Z":  
_mod.use_x = False  
_mod.use_y = False  
_mod.use_z = True  
tion at the end -add  
select= 1  
b.select=1  
t.scene.objects.active  
ected" + str(modifier  
r_ob.select = 0  
.context.selected_obj  
objects[one.name].select  
"please select exactly  
PERATOR CLASSES -----  
s.Operator):  
mirror to the selected  
t.mirror_mirror_x"  
X"
```

How-to headings are common for instructions

Purpose of the instructions

Clarification of the conditions to which the instructions apply

Typographic conventions

Start of the instructions

The typographic conventions act as shorthand

The graphic provides readers with visual confirmation that they are doing things correctly.

How to Change the Size or Resolution of Internet Graphics for Print Resolution

The following instructions show you how to change an Internet graphic from screen resolution to print resolution.

Internet graphics are optimized for quick download by having their resolution reduced to 72 dpi (dots per inch), which is the resolution at which monitors display information. However, when you download Internet graphics and print them (for instance in a report) the 72 dpi resolution will look very pixilated. You, therefore, need to change the resolution of Internet graphics to 150 dpi before inserting them into documents you intend to print on paper.

These instructions assume that you will be using Adobe Photoshop CS, the digital editing software available in your computer labs in NE1. Other digital editing software packages offer the same functions, but call them by different names and sort them differently in their menu systems.

Conventions

These instructions use the following typographic conventions:

Dialog Window Buttons and Keyboard Commands:

Menu selection sequences:

Dialog Windows

Copy, Paste, Save As, Okay

[Start] > [Programs] > [Adobe] > [Adobe Photoshop CS]

"New File Dialog Window"

Changing the resolution of a graphic

When you find an Internet graphic you want to use in your report, please do the following:

1. Right click on the graphic and select Copy from the "Windows Pop-up Menu."



2. Open Adobe Photoshop CS: [Start] > [Programs] > [Adobe] > [Adobe Photoshop CS].

Adobe Photoshop is a memory intensive program and will take some time to load. Once in Photoshop,

```
object to mirror  
_mod.mirror_object  
ion == "MIRROR_X":  
_mod.use_x = True  
_mod.use_y = False  
_mod.use_z = False  
ation == "MIRROR_Y":  
_mod.use_x = False  
_mod.use_y = True  
_mod.use_z = False  
ation == "MIRROR_Z":  
_mod.use_x = False  
_mod.use_y = False  
_mod.use_z = True  
tion at the end -add  
select= 1  
b.select=1  
t.scene.objects.active  
ected" + str(modifier  
r_ob.select = 0  
.context.selected_obj  
objects[one.name].select  
"please select exactly  
PERATOR CLASSES -----  
s.Operator):  
mirror to the selected  
t.mirror_mirror_x"  
X"
```

Consistent, parallel use of “at this point” makes the instructions easier to understand. It signals that the following information indicates what happens next; it’s a check to see that the reader has performed the step correctly.

Always look for ways to make your instructions more concise (and clear) and more consistent.

3. Open a new Photoshop file: [File] > [New].

At this point, the Photoshop “New File Dialog Window” will open. Please note that Photoshop will automatically create a file that matches the size and resolution of the graphic in the Windows Pasteboard.



4. Change the default file name to a file name that will make it easy for you to identify the contents of the file.

5. Click Okay.

At this point, you will see a blank file. Photoshop is waiting for you to input content.

6. Paste the contents of your Windows Pasteboard into the Photoshop file: Ctrl + V (paste command).



At this point, you should see your image in the file window. You now need to change the resolution of this image.

Usability

Procedures aren't any good if the intended reader cannot follow them.

No. 1 mistake of technical writers (or SMEs): assume that the reader knows as much as the writer.

Form a clear picture of your reader.

Understand that they are reaching for your instructions precisely because they don't know what to do.



COIK

clear only if known

Planning for Usability

Who are your readers? What is their technical know-how? Relevant skills?

Experienced users or fellow coders tasked with a procedure will bring more skills to the task than the general population.

Clearly analyze what tasks the reader will need to perform.

Figure out what level of detail to provide.

```
mirror_mod.mirror_object = mirror_obj
```

```
operation == "MIRROR_X":  
    mirror_mod.use_x = True
```

```
    mirror_mod.use_y = False  
    mirror_mod.use_z = False
```

```
operation == "MIRROR_Y":  
    mirror_mod.use_x = False
```

```
    mirror_mod.use_y = True  
    mirror_mod.use_z = False
```

```
operation == "MIRROR_Z":  
    mirror_mod.use_x = False
```

```
    mirror_mod.use_y = False  
    mirror_mod.use_z = True
```

```
selection at the end -add  
mirror_ob.select= 1
```

```
mirror_ob.select=1  
context.scene.objects.active
```

```
obj = ("Selected" + str(modifier.name))  
mirror_ob.select = 0
```

```
obj = bpy.context.selected_objects[0]  
data.objects[one.name].select
```

```
print("please select exactly one object")
```

```
----- OPERATOR CLASSES -----
```

```
class MirrorX(bpy.types.Operator):
```

```
    """Mirror X mirror to the selected object"""
```

```
    bl_label = "Mirror X"
```

```
    bl_options = {'DEFAULT', 'REGISTERED'}  
    mirror_x = True
```

```
    def execute(self, context):  
        mirror_obj = context.selected_objects[0]
```

Testing for Usability

Find someone like your target reader to perform the procedure.

Find someone who has never performed the procedure before and has little knowledge of it.

Don't help them; see how they do.

Take note where they get confused or stuck; clarify those parts.

Retest, using different testers.

```
mirror object to mirror  
mirror_mod.mirror_object
```

```
operation == "MIRROR_X":  
mirror_mod.use_x = True  
mirror_mod.use_y = False  
mirror_mod.use_z = False
```

```
operation == "MIRROR_Y":  
mirror_mod.use_x = False  
mirror_mod.use_y = True  
mirror_mod.use_z = False
```

```
operation == "MIRROR_Z":  
mirror_mod.use_x = False  
mirror_mod.use_y = False  
mirror_mod.use_z = True
```

```
selection at the end -add  
mirror_ob.select= 1  
mirror_ob.select=1
```

```
context.scene.objects.active  
("Selected" + str(modifier))  
mirror_ob.select = 0  
= bpy.context.selected_object  
data.objects[one.name].select
```

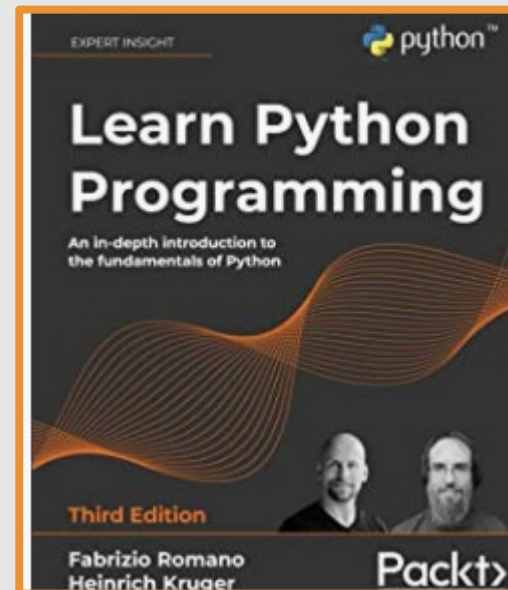
```
print("please select exactly  
-- OPERATOR CLASSES --
```

```
types.Operator):  
X mirror to the selected  
object.mirror_mirror_x"  
mirror X"
```


Manuals

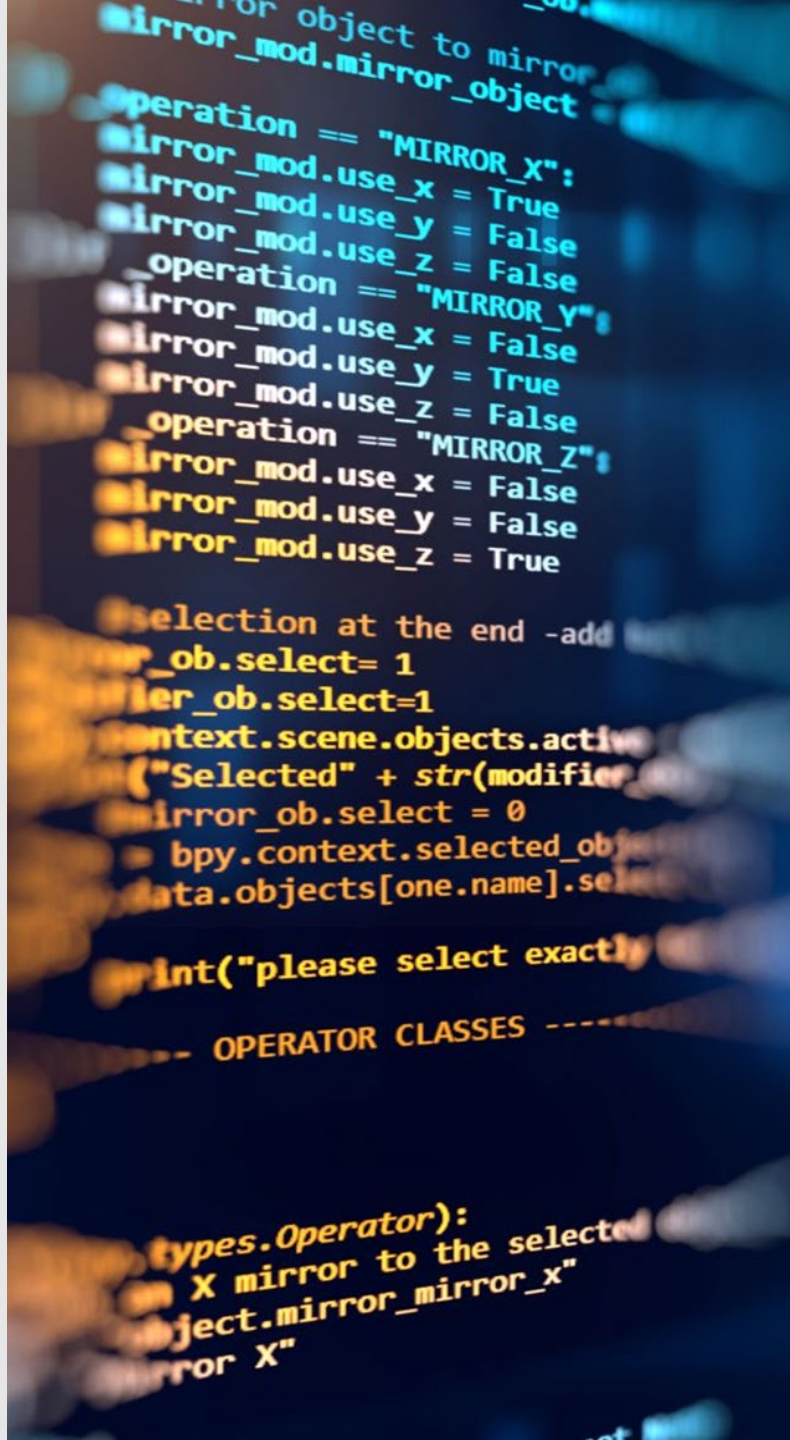
Bring together multiple procedures for a common purpose:

- A software manual brings together procedures for all the tasks the software can perform.
- A bike maintenance manual brings together procedures for maintaining all the systems of a bike.



Planning and writing manuals

1. Task analysis: figure out who will use the manual and what they will need/want to do.
2. Group and organize tasks into a logical order:
 - Most to least frequently performed procedure
 - Most to least important procedure
 - Chronological sequence
 - Other (some order will generally present itself)
3. Write each procedure in a style and level of detail appropriate to the reader.
4. Provide a thorough introduction to the manual; this will contain an overall description (definition) of the mechanism or software. But each procedure within the manual will also include an introduction to that procedure.



Sample Manual Outline

Introduction

1. Tools: putting together a tool kit
2. Basics: pre-ride inspection, wheel removal, general cleaning
3. Emergency repairs
4. Chains
5. Transmissions
6. Brakes
7. Pedals and wheels
8. Cranks and bottom brackets
9. Handlebars, stems, headsets
10. Wheel building
11. Forks and frames

To help with the planning of your instructions:

- Download the plan sheet from D2L
- Carefully think about every step that a reader will have to perform
- Consider including a graphic that would help
- Consider whether a novice reader will require some additional information to perform the step correctly or to feel reassured that they have performed a step correctly

Walkthrough the instructions

Use this table before you write the instructions. Walk through the software and write down every step required. Note down any specific graphics needed. **Do not insert the graphics**, just note down what the user should see specifically. Add any other information in the notes area.

[illegible]

Homework for this Week

Lecture

Quiz 12 is open until Nov. 16 at 23:59

In the lab

This week's labs:

1. Form groups (reform the project group in which you created your apps)
2. Sign up on the Sign-up Sheet for Instruction Manuals. Indicate the app you will document
3. Wait for instructor to create group folders
4. Submit link to shared file with Editing access for the instructor



Questions



Thank you

